

Manuel Alejandro Reyes Jiménez

Mathematics · Teaching · Research

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RESEARCH

2026 – present **PhD student in Applied Mathematics**
Universitat Politècnica de Catalunya (UPC) — FME
Collatz conjecture and symbolic dynamics. Advisor: Dr. Jaume Franch.

TEACHING EXPERIENCE

2018 – present **Mathematics teacher**
Col·legi Maristes Sants—Les Corts (Barcelona)
IB Mathematics AI HL and SL, Bachillerato Mathematics, Mathematics 2nd ESO. Cangur preparation. ICT Coordinator (2019–2024).

2011 – 2018 **Mathematics teacher · Head of Studies (Bachillerato)**
Escola Meritxell (Mataró)
ESO and Bachillerato mathematics. Last two years as Head of Studies for Bachillerato.

PUBLICATIONS

2026 **A Fibonacci theorem for Collatz trajectories via modular graph structure**
A. Reyes — preprint, arXiv:2606.02621 [math.NT]

2014 **New conditions to keep static feedback linearizability under addition of new inputs**
J. Franch, A. Reyes — 53rd IEEE Conference on Decision and Control (CDC), pp. 5624–5629

2013 **Differential flatness of a class of n-DOF planar manipulators driven by an arbitrary number of actuators**
J. Franch, A. Reyes, S. K. Agrawal — European Control Conference (ECC), pp. 161–166

EDUCATION

2010 – 2012 **MSc in Applied Mathematics**
Universitat Politècnica de Catalunya — FME

2010 – 2011 **MSc in Teacher Training (Mathematics)**
Universitat Politècnica de Catalunya — FME

2004 – 2010 **BSc in Mathematics (Licenciatura)**
Facultat de Matemàtiques i Estadística (FME) — UPC

LANGUAGES

Native **Spanish · Catalan**

Intermediate **English**
B2 level. 9-month Erasmus exchange at the University of Sheffield.

TOOLS & SOFTWARE

Moodle · ClickEdu · Alexia · LaTeX · GeoGebra · Maple · Matlab · C++ · Office 365 · Notion · Generative AI (LLMs)